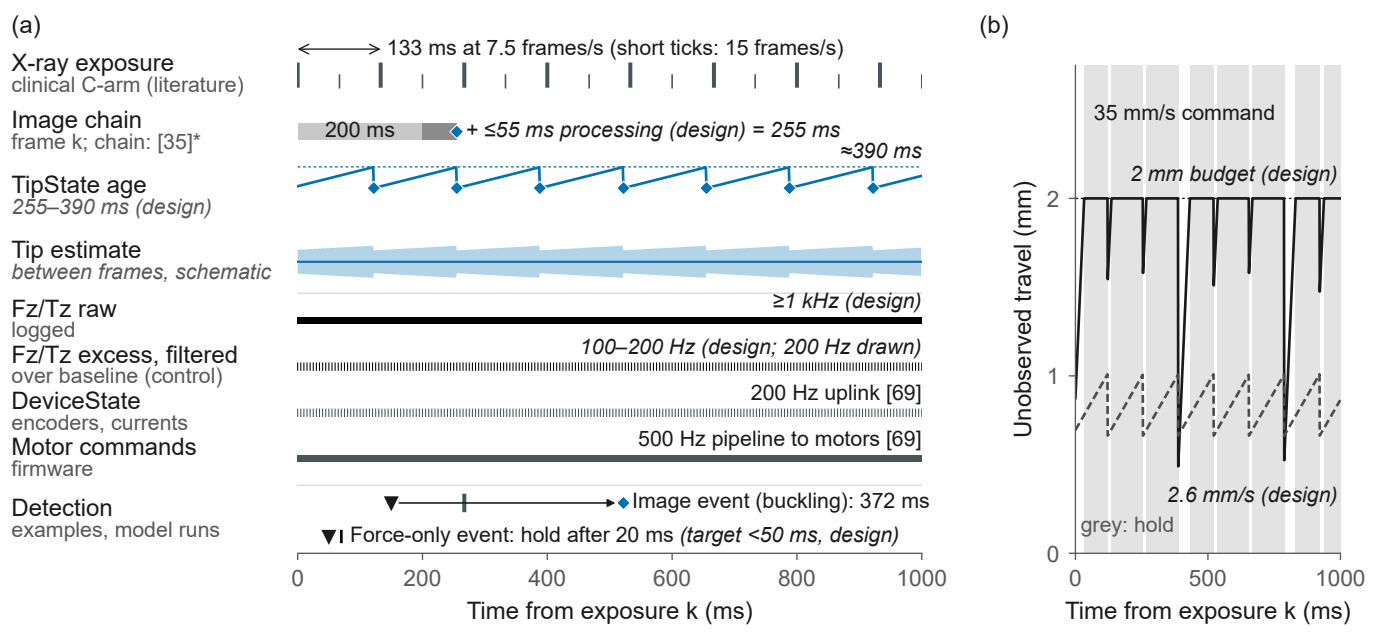

(a)
X-ray exposure
clinical C-arm (literature)
133 ms at 7.5 frames/s (short ticks: 15 frames/s)
Image chain
frame k; chain: [35]*
200 ms
+ ≤55 ms processing (design) = 255 ms
≈390 ms
TipState age
255–390 ms (design)
Tip estimate
between frames, schematic
Fz/Tz raw
logged
≥1 kHz (design)
Fz/Tz excess, filtered
over baseline (control)
100–200 Hz (design; 200 Hz drawn)
DeviceState
encoders, currents
200 Hz uplink [69]
Motor commands
firmware
500 Hz pipeline to motors [69]
Detection
examples, model runs
Image event (buckling): 372 ms
Force-only event: hold after 20 ms (target <50 ms, design)
0
200
400
600
800
1000
Time from exposure k (ms)
(b)
35 mm/s command
2 mm budget (design)
2.6 mm/s (design)
grey: hold
Unobserved travel (mm)
0
1
2
0
500
1000
Time from exposure k (ms)